

Ph.D. Qualifying Exam 2025–Theory

July 8 2025
9:30am–2:30pm

Rules: This is a five-hour exam consisting of FOUR PARTS with equal weights. PART I, II and III are closed notes and closed book. For PART IV, you may use ONE page (front and back, 8.5 by 11 inch paper) “information sheet” for Stat 594. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly. Please put your name or initials on every page. Finally, write your answers clearly and legibly in the space provided after each problem, preferably using pens, leaving an impression deep enough for the papers to be scanned.

Part I

Instructions:

- Please only write down correct and concise arguments.
- Points will be taken off for any incorrect and redundant arguments.

Problem. Suppose $X_1, X_2, \dots$ are independent and $S_n \equiv \sum_{m=1}^n X_m$.

- (i) Use Lindeberg-Feller central limit theorem to show that if $\sup_m |X_m| < \infty$ a.s., and $\sum_m \text{Var}(X_m) = \infty$, then $(S_n - \mathbb{E}S_n)/\sqrt{\text{Var}(S_n)} \xrightarrow{d} \mathcal{N}(0, 1)$.

- (ii) Construct an explicit example such that the CLT holds for S_n with $(S_n - \mathbb{E}S_n)/\sqrt{n} \xrightarrow{d} \mathcal{N}(0, 1)$, but $\text{Var}(S_n)/n$ does not converge to 1.

Part II

Instructions:

- Please only write down correct and concise arguments.
- Points will be taken off for any incorrect and redundant arguments.

Problem. Let $S_n = \sum_{i=1}^n \xi_i$ be an asymmetric simple random walk starting from 0 with $1/2 < p < 1$. Let $T_1 \equiv \inf\{n : S_n = 1\}$.

(i) Construct an appropriate martingale to show that

$$e^\theta \mathbb{E} \phi(\theta)^{-T_1} = 1$$

holds for all $\theta \geq 0$, where $\phi(\theta) = \mathbb{E} \exp(\theta \xi_1) = pe^\theta + qe^{-\theta}$.

(ii) Use (i) to compute $\mathbb{E}s^{T_1}$ for $s \in (0, 1)$.

Part III

Please read the following general instructions carefully.

1. This part is closed-book and closed-note.
2. Please work on your solutions **independently**.
3. Please show the details of your derivations.
4. Partial credit will be given if you make reasonable progress but cannot solve the problem completely.

Problem 1. MLE and Hypothesis Testing for Exponential Family. (25 points)

Consider that we have access to i.i.d. data $X_1, \dots, X_n$ following a one-parameter exponential family distribution:

$$p(x \mid \theta) = h(x) \exp \{ \eta(\theta) T(x) - B(\theta) \}, \quad \text{for } x \in \mathcal{X},$$

where $\mathcal{X}$ denotes the support. Assume that $\eta(\theta)$ is a monotonically increasing function of θ , and that $\sum_{i=1}^n T(X_i)$ is a continuous random variable.

- (i) Construct the uniformly most powerful (UMP) test for the hypotheses $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.

(ii) Prove that the test constructed in part (i) is uniformly most powerful.

- (iii) Prove or disprove the existence of a uniformly most powerful test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Problem 2. Efficiency Bound and One-step Estimator. (50 points)

Consider that $X_1, \dots, X_n$ are i.i.d. with density function $p_\theta(\cdot)$. Let $l_n(\theta)$ denote the negative log-likelihood function, and let $H_n(\theta)$ denote its Hessian. Assume that the required regularity conditions are satisfied.

- (i) Suppose our goal is to estimate $\psi(\theta)$, where $\psi(\cdot)$ is a function of θ . State the Cramér-Rao lower bound for any unbiased estimator $T(X)$ satisfying $\mathbb{E}[T(X)] = \psi(\theta)$. Provide a proof for your statement.

- (ii) Suppose $\tilde{\theta}$ is an initial estimator of θ satisfying $\|\tilde{\theta} - \theta\|_2 = o_p(n^{-1/4})$, and that $H_n(\theta)$ is L -Lipschitz. Construct the Newton-Raphson estimator $\hat{\theta}$ and show that it is asymptotically normal and achieves the Cramér-Rao lower bound.
(Hint: In your proof, you may assume that the MLE is asymptotically normal and achieves the Cramér-Rao lower bound.)

Problem 3. Reweighting Estimator for Distributional Shift. (25 points)

Let $\{(X_i, Y_i, R_i)\}_{i=1}^n$ be i.i.d. observations, where $X_i \in \mathbb{R}^d$ denotes covariates, $Y_i \in \mathbb{R}$ denotes a response, and $R_i \in \{0, 1\}$ is an indicator such that $R_i = 1$ if Y_i is observed and $R_i = 0$ if Y_i is missing. Assume the following:

- (a) The conditional distribution of the response given covariates does not depend on the outcome missingness indicator:

$$Y_i \mid X_i, R_i = 1 \stackrel{d}{=} Y_i \mid X_i, R_i = 0.$$

- (b) The marginal distribution of the covariates differs between observed and missing responses:

$$X_i \mid R_i = 1 \sim P_X, \quad X_i \mid R_i = 0 \sim Q_X,$$

where P_X and Q_X may differ.

We aim to estimate the target parameter

$$\beta^* = \arg \min_{\beta \in \mathbb{R}^d} \mathbb{E} \left[(Y_i - X_i^\top \beta)^2 \mid R_i = 0 \right],$$

- (i) Show that

$$\mathbb{E} \left[(Y_i - X_i^\top \beta)^2 \mid R_i = 0 \right] = \mathbb{E} \left[(Y_i - X_i^\top \beta)^2 \cdot \frac{f_{X|R=0}(X_i)}{f_{X|R=1}(X_i)} \mid R_i = 1 \right],$$

where $f_{X|R=r}(\cdot)$ denotes the conditional density of X_i given $R_i = r$.

(ii) Prove that

$$\frac{f_{X|R=0}(x)}{f_{X|R=1}(x)} = \frac{\mathbb{P}(R=0 \mid x)}{\mathbb{P}(R=1 \mid x)} \cdot \frac{\mathbb{P}(R=1)}{\mathbb{P}(R=0)}.$$

(iii) Apply (i) and (ii) to construct an estimator of β^* .

Part IV

Suppose that (X_i, Y_i) , $i = 1, \dots, n$, are independent and identically distributed observations as (X, Y) , satisfying a linear regression model

$$E(Y|X) = X^T \beta^*,$$

where β^* is a $p \times 1$ vector of unknown parameters. Split $X = (Z, W^T)^T$ and $\beta^* = (\theta^*, \alpha^{*T})^T$, where Z and θ^* are scalars. Let $\hat{\beta} = (\hat{\theta}, \hat{\alpha}^T)^T$ be a Lasso-penalized least-squares estimator of β^* :

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2n} \sum_{i=1}^n (Y_i - X_i^T \beta)^2 + \lambda \|\beta\|_1 \right\}, \quad (1)$$

where $\lambda > 0$ is a tuning parameter. Suppose in addition that Z is a binary covariate (taking value 0 or 1). Let $\hat{\gamma}$ be a Lasso-penalized maximum likelihood estimator in a logistic regression model of Z_i on W_i , $i = 1, \dots, n$:

$$\hat{\gamma} = \operatorname{argmin}_{\gamma \in \mathbb{R}^{p-1}} \left[\frac{1}{n} \sum_{i=1}^n \{-Z_i W_i^T \gamma + \log(1 + e^{W_i^T \gamma})\} + \lambda_\gamma \|\gamma\|_1 \right], \quad (2)$$

where $\lambda_\gamma > 0$ is a tuning parameter.

Consider a (new) debiased Lasso estimator for θ^* :

$$\tilde{\theta} = \frac{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\hat{\gamma}))^T (\mathbf{Y} - \mathbf{W}\hat{\alpha})/n}{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\hat{\gamma}))^T \mathbf{Z}/n}, \quad (3)$$

where $\mathbf{Y} = (Y_1, \dots, Y_n)^T$, $\mathbf{Z} = (Z_1, \dots, Z_n)^T$, $\mathbf{W} = (W_1^T, \dots, W_n^T)^T$, and $\operatorname{expit}(u) = e^u / (1 + e^u)$ and $\operatorname{expit}(\cdot)$ can be applied componentwise to a vector. It is of interest to obtain an asymptotic expansion for $\tilde{\theta}$ in the form

$$\tilde{\theta} - \theta^* = \frac{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\bar{\gamma}))^T (\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\alpha^*)/n}{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\bar{\gamma}))^T \mathbf{Z}/n} + o_p(n^{-1/2}),$$

where $\bar{\gamma} = \operatorname{argmin}_{\gamma \in \mathbb{R}^{p-1}} E\{-ZW^T \gamma + \log(1 + e^{W^T \gamma})\}$.

From the definition (3), direct calculation gives

$$\tilde{\theta} - \theta^* = \frac{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\hat{\gamma}))^T (\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\hat{\alpha})/n}{(\mathbf{Z} - \operatorname{expit}(\mathbf{W}\hat{\gamma}))^T \mathbf{Z}/n}.$$

The following questions are designed to establish

$$\begin{aligned} & (\mathbf{Z} - \operatorname{expit}(\mathbf{W}\hat{\gamma}))^T (\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\hat{\alpha})/n \\ &= (\mathbf{Z} - \operatorname{expit}(\mathbf{W}\bar{\gamma}))^T (\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\alpha^*)/n + o_p(n^{-1/2}) \end{aligned} \quad (4)$$

under suitable conditions.

For each question, you may use the results from the preceding questions, whether or not you have answered those questions.

1) Show that

$$\begin{aligned}
& |(\mathbf{Z} - \text{expit}(\mathbf{W}\hat{\gamma}))^T(\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\hat{\alpha})/n - (\mathbf{Z} - \text{expit}(\mathbf{W}\bar{\gamma}))^T(\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\alpha^*)/n| \\
& \leq \underbrace{|(\mathbf{Z} - \text{expit}(\mathbf{W}\hat{\gamma}))^T \mathbf{W}(\hat{\alpha} - \alpha^*)/n|}_{\text{I}} \\
& \quad + \underbrace{|(\text{expit}(\mathbf{W}\hat{\gamma}) - \text{expit}(\mathbf{W}\bar{\gamma}))^T(\mathbf{Y} - \mathbf{Z}\theta^* - \mathbf{W}\alpha^*)/n|}_{\text{II}}.
\end{aligned}$$

2) Show that under suitable conditions,

$$\mathbf{I} = O_p(1)(|S_{\beta^*}| + |S_{\bar{\gamma}}|) \frac{\log(n \vee p)}{n}.$$

State explicitly your conditions, which may be in terms of standard properties of $\hat{\beta}$ and $\hat{\gamma}$, for example,

$$\begin{aligned} \|\mathbf{X}^T(\hat{\beta} - \beta^*)\|_2^2/n &= O_p(1)|S_{\beta^*}| \frac{\log(n \vee p)}{n}, \quad \|\hat{\beta} - \beta^*\|_1 = O_p(1)|S_{\beta^*}| \sqrt{\frac{\log(n \vee p)}{n}}, \\ \|A(\bar{\gamma})\mathbf{W}^T(\hat{\gamma} - \bar{\gamma})\|_2^2/n &= O_p(1)|S_{\bar{\gamma}}| \frac{\log(n \vee p)}{n}, \quad \|\hat{\gamma} - \bar{\gamma}\|_1 = O_p(1)|S_{\bar{\gamma}}| \sqrt{\frac{\log(n \vee p)}{n}}, \end{aligned}$$

where $A(\gamma) = \text{diag}(\text{expit}_2(\mathbf{W}\gamma))$, the diagonal matrix with $\text{expit}_2(\mathbf{W}\gamma)$ on the diagonal. Throughout, $\text{expit}_2(u) = \frac{d}{du} \text{expit}(u) = \text{expit}(u)(1 - \text{expit}(u))$, and $\text{expit}_2(\cdot)$ can be applied componentwise to a vector.

3) Show that for any $u, \tilde{u} \in \mathbb{R}$,

$$|\text{expit}_2(u) - \text{expit}_2(\tilde{u})| \leq \text{expit}_2(\tilde{u})|u - \tilde{u}|,$$

and

$$|\text{expit}(u) - \text{expit}(\tilde{u}) - \text{expit}_2(\tilde{u})(u - \tilde{u})| \leq \text{expit}_2(\tilde{u})(u - \tilde{u})^2.$$

4) Assume that $Y - Z\theta^* - W^T\alpha^*$ is bounded by a constant B_0 almost surely. Show that under suitable conditions,

$$\mathbf{II} = O_p(1)(|S_{\beta^*}| + |S_{\tilde{\gamma}}|) \frac{\log(n \vee p)}{n}.$$

State explicitly your conditions.

Remark. Combining Questions 1), 2) and 4) gives the desired result (4) provided that $(|S_{\beta^*}| + |S_{\tilde{\gamma}}|) \frac{\log(n \vee p)}{\sqrt{n}} = o(1)$. [This remark accentuates the conclusion of what you have done, it is not part of the problem which you need to show.]